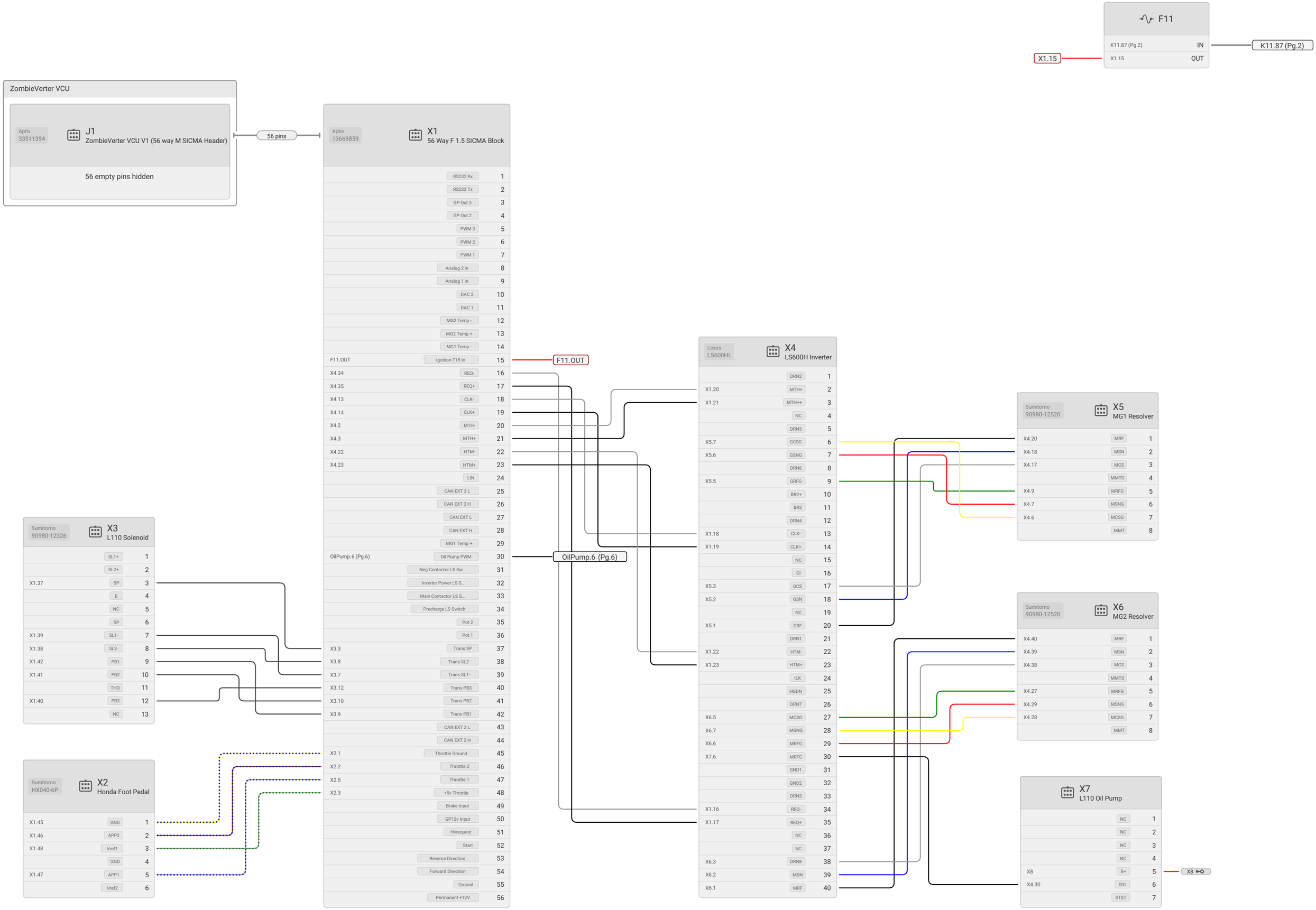
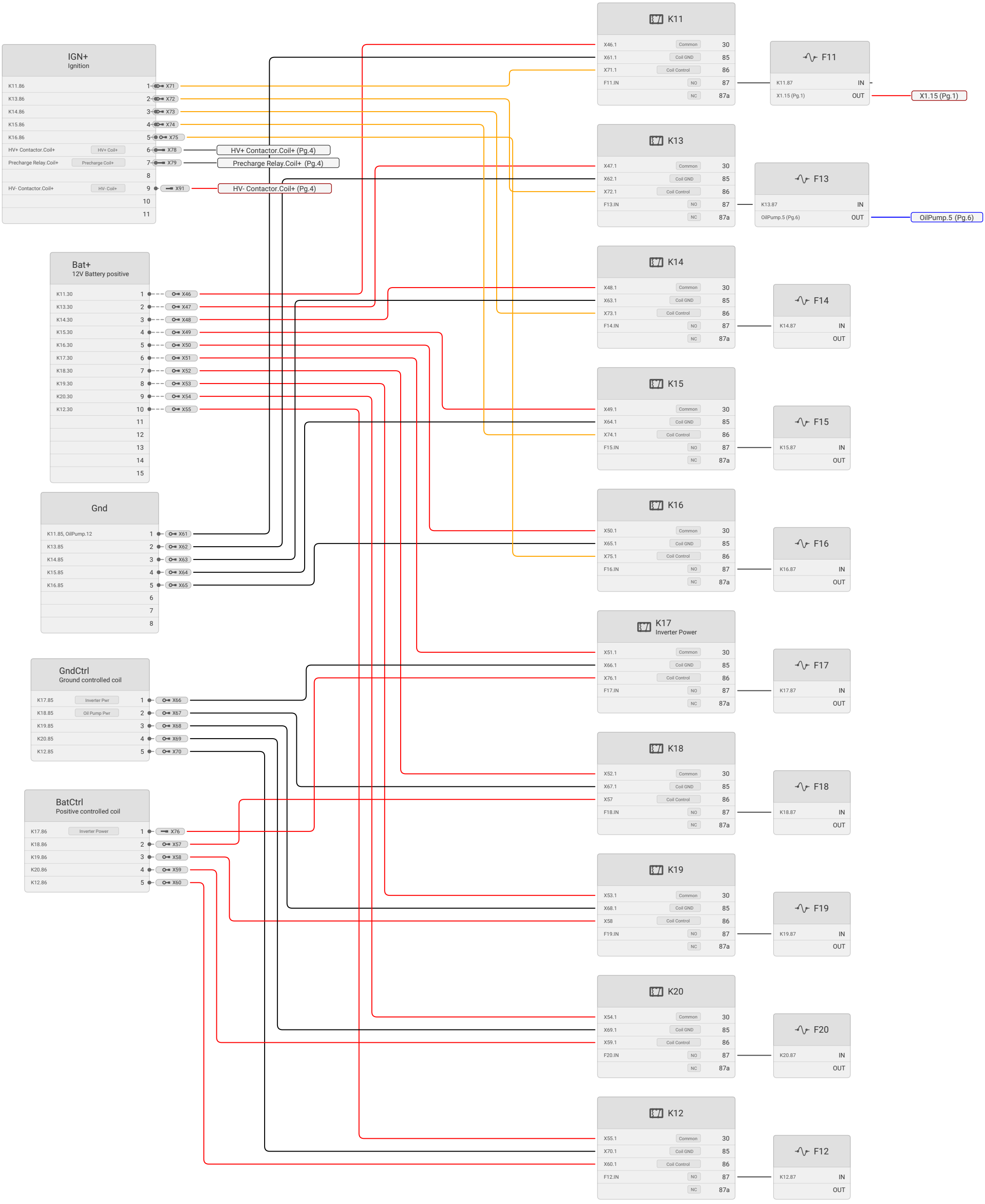
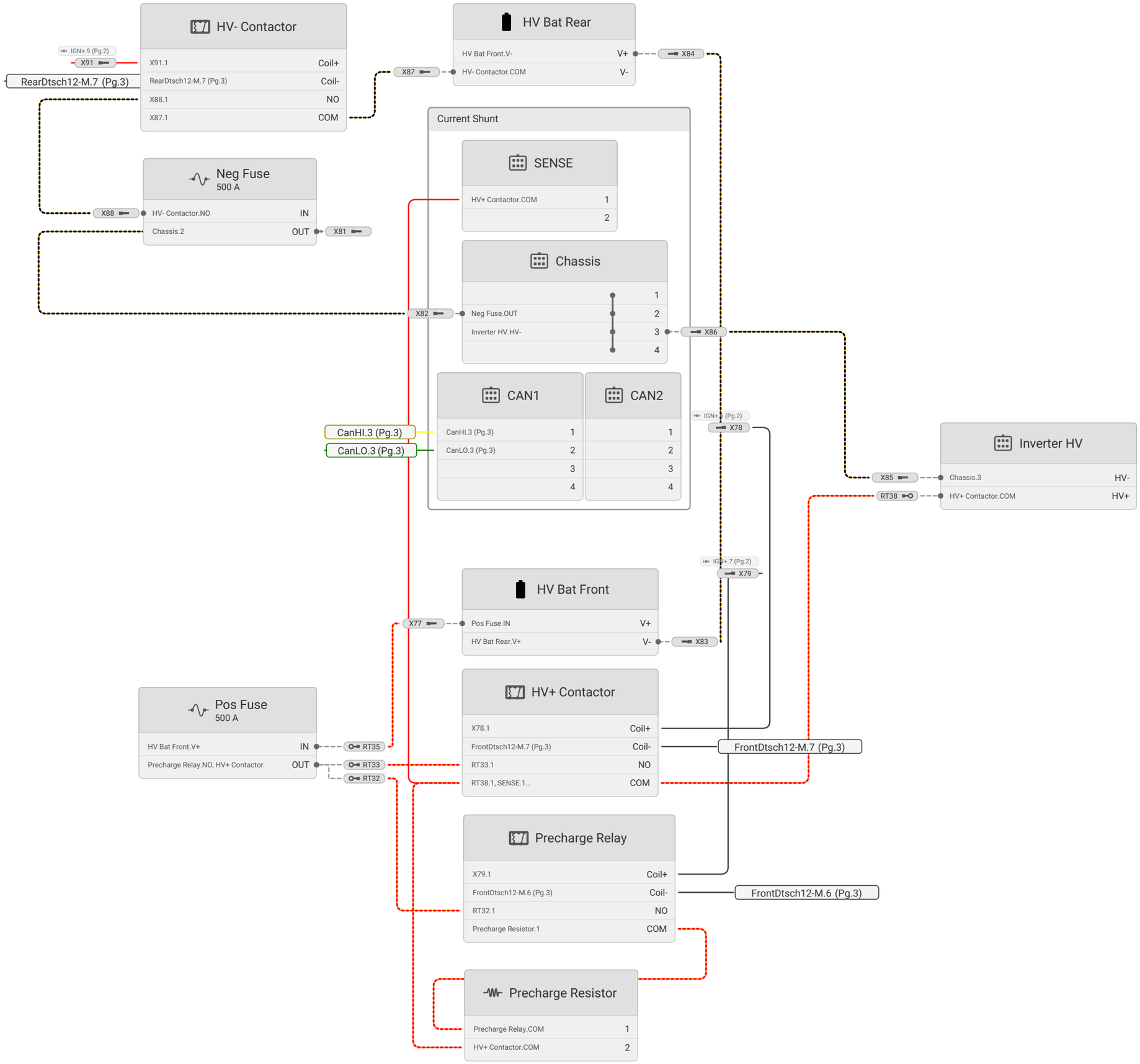
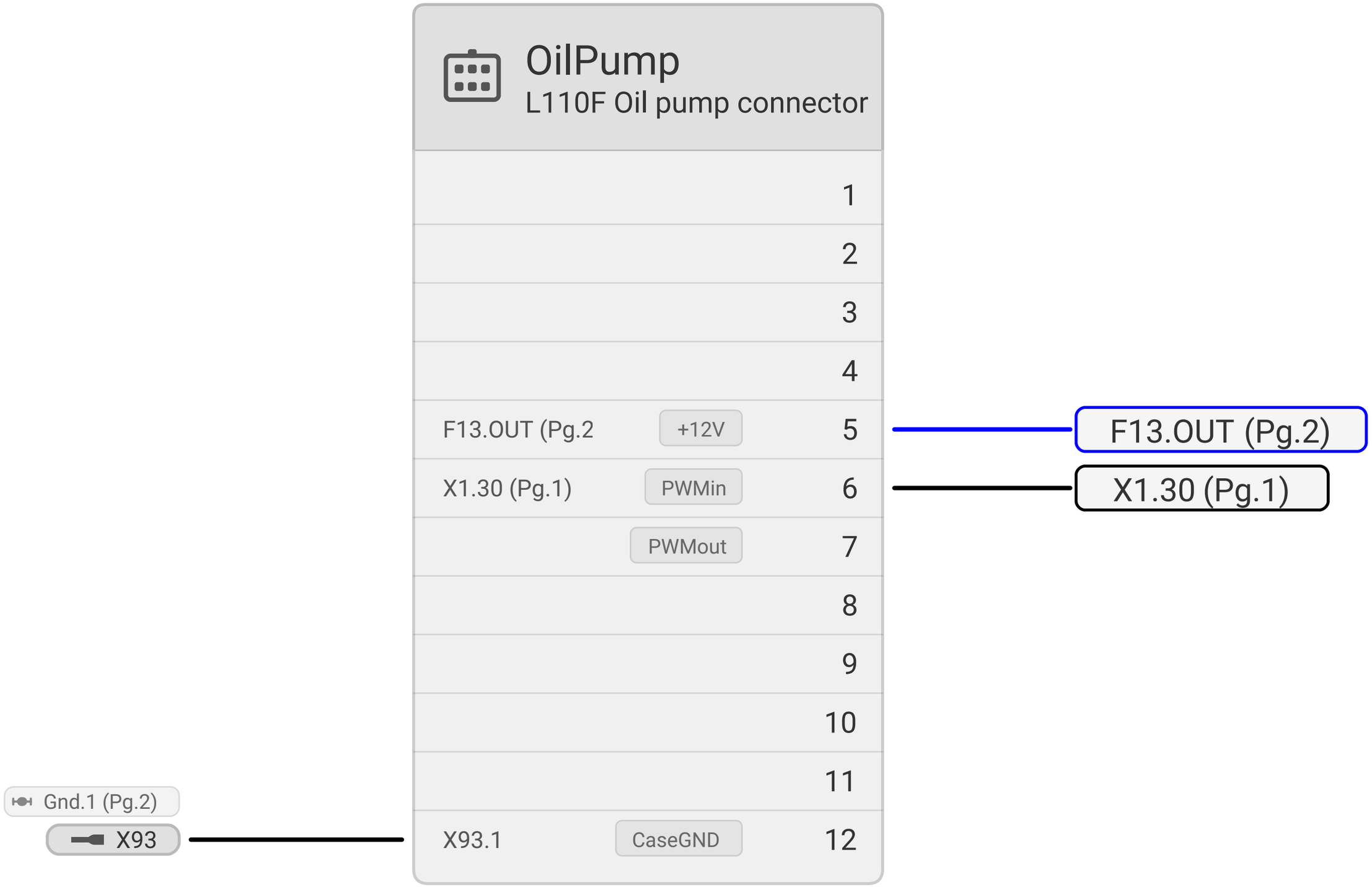

F11
K11.87 (Pg.2) IN
X1.15 OUT
X1.15
K11.87 (Pg.2)
ZombieVerter VCU
Aptiv 33511394
J1
ZombieVerter VCU V1 (56 way M SICMA Header)
56 empty pins hidden
56 pins
Aptiv 13669859
X1
56 Way F 1.5 SICMA Block
1 RS232 Rx
2 RS232 Tx
3 GP Out 3
4 GP Out 2
5 PWM 3
6 PWM 2
7 PWM 1
8 Analog 2 in
9 Analog 1 in
10 DAC 2
11 DAC 1
12 MG2 Temp -
13 MG2 Temp +
14 MG1 Temp -
F11.OUT 15 Ignition T15 In
F11.OUT
X4.34 16 REQ-
X4.35 17 REQ+
X4.13 18 CLK-
X4.14 19 CLK+
X4.2 20 MTH-
X4.3 21 MTH+
X4.22 22 HTM-
X4.23 23 HTM+
24 LIN
25 CAN EXT 3 L
26 CAN EXT 3 H
27 CAN EXT L
28 CAN EXT H
29 MG1 Temp +
OilPump.6 (Pg.6) 30 Oil Pump PWM
OilPump.6 (Pg.6)
31 Neg Contactor LS Sw...
32 Inverter Power LS S...
33 Main Contactor LS S...
34 Precharge LS Switch
35 Pot 2
36 Pot 1
X3.3 37 Trans SP
X3.8 38 Trans SL2-
X3.7 39 Trans SL1-
X3.12 40 Trans PB3
X3.10 41 Trans PB2
X3.9 42 Trans PB1
43 CAN EXT 2 L
44 CAN EXT 2 H
X2.1 45 Throttle Ground
X2.2 46 Throttle 2
X2.5 47 Throttle 1
X2.3 48 +5v Throttle
49 Brake Input
50 GP12v Input
51 Hvrequest
52 Start
53 Reverse Direction
54 Forward Direction
55 Ground
56 Permanent +12V
Sumitomo 90980-12326
X3
L110 Solenoid
1 SL1+
2 SL2+
X1.37 3 SP
4 E
5 NC
6 SP
X1.39 7 SL1-
X1.38 8 SL2-
X1.42 9 PB1
X1.41 10 PB2
11 THO
X1.40 12 PB3
13 NC
Sumitomo HX040-6P
X2
Honda Foot Pedal
X1.45 1 GND
X1.46 2 APP2
X1.48 3 Vref1
4 GND
X1.47 5 APP1
6 Vref2
Lexus LS600HL
X4
LS600H Inverter
1 DRN2
X1.20 2 MTH+
X1.21 3 MTH++
4 NC
5 DRN5
X5.7 6 GCSG
X5.6 7 GSNG
8 DRN6
X5.5 9 GRFG
10 BR2+
11 BR2
12 DRN4
X1.18 13 CLK-
X1.19 14 CLK+
15 NC
16 GI
X5.3 17 GCS
X5.2 18 GSN
19 NC
X5.1 20 GRF
21 DRN1
X1.22 22 HTM-
X1.23 23 HTM+
24 ILK
25 HGDN
26 DRN7
X6.5 27 MCSG
X6.7 28 MSNG
X6.6 29 MRFG
X7.6 30 MRFG
31 GND1
32 GND2
33 DRN3
X1.16 34 REQ-
X1.17 35 REQ+
36 NC
37 NC
X6.3 38 DRN8
X6.2 39 MSN
X6.1 40 MRF
Sumitomo 90980-12520
X5
MG1 Resolver
X4.20 1 MRF
X4.18 2 MSN
X4.17 3 MCS
4 MMTD
X4.9 5 MRFG
X4.7 6 MSNG
X4.6 7 MCSG
8 MMT
Sumitomo 90980-12520
X6
MG2 Resolver
X4.40 1 MRF
X4.39 2 MSN
X4.38 3 MCS
4 MMTD
X4.27 5 MRFG
X4.29 6 MSNG
X4.28 7 MCSG
8 MMT
X7
L110 Oil Pump
1 NC
2 NC
3 NC
4 NC
X8 5 B+
X8
X4.30 6 SIG
7 STST


IGN+
Ignition
K11.86 1 X71
K13.86 2 X72
K14.86 3 X73
K15.86 4 X74
K16.86 5 X75
HV+ Contactor.Coil+ HV+ Coil+ 6 X78
Precharge Relay.Coil+ Precharge Coil+ 7 X79
8
HV- Contactor.Coil+ HV- Coil+ 9 X91
10
11
HV+ Contactor.Coil+ (Pg.4)
Precharge Relay.Coil+ (Pg.4)
HV- Contactor.Coil+ (Pg.4)
Bat+
12V Battery positive
K11.30 1 X46
K13.30 2 X47
K14.30 3 X48
K15.30 4 X49
K16.30 5 X50
K17.30 6 X51
K18.30 7 X52
K19.30 8 X53
K20.30 9 X54
K12.30 10 X55
11
12
13
14
15
Gnd
K11.85, OilPump.12 1 X61
K13.85 2 X62
K14.85 3 X63
K15.85 4 X64
K16.85 5 X65
6
7
8
GndCtrl
Ground controlled coil
K17.85 Inverter Pwr 1 X66
K18.85 Oil Pump Pwr 2 X67
K19.85 3 X68
K20.85 4 X69
K12.85 5 X70
BatCtrl
Positive controlled coil
K17.86 Inverter Power 1 X76
K18.86 2 X57
K19.86 3 X58
K20.86 4 X59
K12.86 5 X60
K11
X46.1 Common 30
X61.1 Coil GND 85
X71.1 Coil Control 86
F11.IN NO 87
NC 87a
F11
K11.87 IN
X1.15 (Pg.1) OUT
X1.15 (Pg.1)
K13
X47.1 Common 30
X62.1 Coil GND 85
X72.1 Coil Control 86
F13.IN NO 87
NC 87a
F13
K13.87 IN
OilPump.5 (Pg.6) OUT
OilPump.5 (Pg.6)
K14
X48.1 Common 30
X63.1 Coil GND 85
X73.1 Coil Control 86
F14.IN NO 87
NC 87a
F14
K14.87 IN
OUT
K15
X49.1 Common 30
X64.1 Coil GND 85
X74.1 Coil Control 86
F15.IN NO 87
NC 87a
F15
K15.87 IN
OUT
K16
X50.1 Common 30
X65.1 Coil GND 85
X75.1 Coil Control 86
F16.IN NO 87
NC 87a
F16
K16.87 IN
OUT
K17
Inverter Power
X51.1 Common 30
X66.1 Coil GND 85
X76.1 Coil Control 86
F17.IN NO 87
NC 87a
F17
K17.87 IN
OUT
K18
X52.1 Common 30
X67.1 Coil GND 85
X57 Coil Control 86
F18.IN NO 87
NC 87a
F18
K18.87 IN
OUT
K19
X53.1 Common 30
X68.1 Coil GND 85
X58 Coil Control 86
F19.IN NO 87
NC 87a
F19
K19.87 IN
OUT
K20
X54.1 Common 30
X69.1 Coil GND 85
X59.1 Coil Control 86
F20.IN NO 87
NC 87a
F20
K20.87 IN
OUT
K12
X55.1 Common 30
X70.1 Coil GND 85
X60.1 Coil Control 86
F12.IN NO 87
NC 87a
F12
K12.87 IN
OUT

DCDC-R
12-5 DCDC
RearDtsch12-M.1 Gnd 1
RearDtsch12-M.12 +12 2
BMS-R.2 +5 3
BMS-R.1 Gnd 4
BMS-R
BMS Slaves
DCDC-R.4 Gnd 1
DCDC-R.3 +5 2
RearDtsch12-M.4 CanHI 3
RearDtsch12-M.5 CanLo 4
Term
120ohm Termin...
BMS-R.3 1
BMS-R.4 2
RearDtsch12-M
Rear BatBox Signal connector
DCDC-R.1 Gnd 1
CANHi-in 2
CANLO-in 3
BMS-R.3 CANHI-Return 4
BMS-R.4 CANLO-Return 5
PreCharge 6
HV- Contactor.Coil- (Pg.4) Contactor 7
Welded+ 8
Welded- 9
ISA1+ 10
ISA1- 11
DCDC-R.2 BAT+ 12
HV- Contactor.Coil- (Pg.4)
RearDtsch12-F
Rear BatBox Signal conn...
FrontDtsch12 Gnd 1
FrontDtsch12 CANHi-in 2
FrontDtsch12 CANLO-in 3
FrontDtsch12 CANHI-R... 4
FrontDtsch12 CANLO-R... 5
FrontDtsch12 PreChar... 6
FrontDtsch12 Contact... 7
FrontDtsch12 Welded+ 8
FrontDtsch12 Welded- 9
FrontDtsch12 ISA1+ 10
FrontDtsch12 ISA1- 11
FrontDtsch12 BAT+ 12
WG1
FrontDtsch12-F
Front BatBox Signal con...
RearDtsch12 Gnd 1
RearDtsch12 CANHi-in 2
RearDtsch12 CANLO-in 3
RearDtsch12 CANHI-R... 4
RearDtsch12 CANLO-R... 5
RearDtsch12 PreChar... 6
RearDtsch12 Contact... 7
RearDtsch12 Welded+ 8
RearDtsch12 Welded- 9
RearDtsch12 ISA1+ 10
RearDtsch12 ISA1- 11
RearDtsch12 BAT+ 12
FrontDtsch12-M
Rear BatBox Signal connector
DCDC-F.1 Gnd 1
CANHi-in 2
CANLO-in 3
CanHI.1 CANHI-Return 4
CanLO.1 CANLO-Return 5
Precharge Relay.Coil- (Pg.4) PreCharge 6
HV+ Contactor.Coil- (Pg.4) Contactor 7
Welded+ 8
Welded- 9
ISA1+ 10
ISA1- 11
DCDC-F.2 BAT+ 12
DCDC-F
12-5 DCDC
FrontDtsch12-M.1 Gnd 1
FrontDtsch12-M.12 +12 2
BMS-F.2 +5 3
BMS-F.1 Gnd 4
BMS-F
BMS Slaves
DCDC-F.4 Gnd 1
DCDC-F.3 +5 2
CanHI.2 CanHI 3
CanLO.2 CanLo 4
CanHI
CanHI
FrontDtsch12-M.4 1
BMS-F.3 2
CAN1.1 (Pg.4) 3
4
5
6
CanLO
CanLO
FrontDtsch12-M.5 1
BMS-F.4 2
CAN1.2 (Pg.4) 3
4
5
6
HV+ Contactor.Coil- (Pg.4)
Precharge Relay.Coil- (Pg.4)
CAN1.1 (Pg.4)
CAN1.2 (Pg.4)


HV- Contactor
IGN+.9 (Pg.2)
X91
X91.1
Coil+
RearDtsch12-M.7 (Pg.3)
RearDtsch12-M.7 (Pg.3)
Coil-
X88.1
NO
X87.1
COM
HV Bat Rear
HV Bat Front.V-
V+
X84
X87
HV- Contactor.COM
V-
Current Shunt
SENSE
HV+ Contactor.COM
1
2
Neg Fuse
500 A
X88
HV- Contactor.NO
IN
Chassis.2
OUT
X81
Chassis
1
X82
Neg Fuse.OUT
2
Inverter HV.HV-
3
X86
4
CAN1
CAN2
CanHI.3 (Pg.3)
CanLO.3 (Pg.3)
CanHI.3 (Pg.3)
1
1
CanLO.3 (Pg.3)
2
2
3
3
4
4
IGN+.3 (Pg.2)
X78
Inverter HV
X85
Chassis.3
HV-
RT38
HV+ Contactor.COM
HV+
IGN+.7 (Pg.2)
X79
HV Bat Front
X77
Pos Fuse.IN
V+
HV Bat Rear.V+
V-
X83
HV+ Contactor
Pos Fuse
500 A
X78.1
Coil+
HV Bat Front.V+
IN
RT35
FrontDtsch12-M.7 (Pg.3)
FrontDtsch12-M.7 (Pg.3)
Coil-
Precharge Relay.NO, HV+ Contactor
OUT
RT33
RT33.1
NO
RT32
RT38.1, SENSE.1...
COM
Precharge Relay
X79.1
Coil+
FrontDtsch12-M.6 (Pg.3)
FrontDtsch12-M.6 (Pg.3)
Coil-
RT32.1
NO
Precharge Resistor.1
COM
Precharge Resistor
Precharge Relay.COM
1
HV+ Contactor.COM
2


OilPump
L110F Oil pump connector
1
2
3
4
F13.OUT (Pg.2
+12V
5
X1.30 (Pg.1)
PWMin
6
PWMout
7
8
9
10
11
X93.1
CaseGND
12
F13.OUT (Pg.2)
X1.30 (Pg.1)
Gnd.1 (Pg.2)
X93